



Tribhuvan University
Institute of Science and Technology



Himalaya College of Engineering
Chyasal, Lalitpur



Project proposal

SatyaScan – An AI-Based Fact-Checking System

Submitted to:

Department of CSIT

Himalaya College of Engineering, Chyasal, Lalitpur

In Partial Fulfilment of the requirements for the degree of Bachelor of Science, Computer Science and Information Technology(BSc. CSIT)

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1. Introduction

1.1 Introduction

The rapid growth of digital media and online platforms has significantly changed the way information is created, shared, and consumed. News and factual content can now be published and disseminated instantly through websites, blogs, and social media platforms. While this has improved access to information, it has also led to the widespread circulation of misinformation and false claims, commonly referred to as fake news. Such content can influence public opinion, create social panic, and negatively impact decision-making processes.

Traditional fact-checking methods rely heavily on manual verification by journalists and domain experts, which is time-consuming and not scalable for the massive volume of content generated daily. As a result, automated and semi-automated fact-checking systems have gained attention in both academic research and industry applications. These systems aim to analyze textual claims, compare them with reliable information sources, and determine their credibility in an efficient and systematic manner.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have made it possible to process large amounts of textual data and identify patterns related to misinformation. Natural Language Processing (NLP) techniques can be used to extract claims, analyze linguistic features, and compare statements against trusted sources. However, many existing systems depend entirely on black-box AI models or external APIs, which raises concerns regarding transparency, explainability, and academic contribution.

This project proposes the development of an AI-based fact-checking system that incorporates machine learning-based credibility scoring. The system focuses on evaluating the trustworthiness of textual claims by analyzing features such as source reliability, content consistency, and evidence agreement. Instead of relying solely on external AI services, the system emphasizes algorithmic logic and machine learning models designed by the developers, ensuring transparency and academic validity. External AI tools, when used, are limited to supportive tasks such as information retrieval and explanation generation.

1.2 Problem Statement

The increasing volume of online information has made it difficult for individuals to distinguish between verified facts and misleading or false content. Manual fact-checking processes are not sufficient to handle the scale and speed at which misinformation spreads. Additionally, many existing automated fact-checking tools suffer from one or more of the following limitations:

- Heavy dependence on proprietary or black-box AI systems with limited explainability
- Lack of clear credibility scoring mechanisms
- Inability to justify verification decisions in a transparent manner
- Limited academic focus on algorithmic design and evaluation

Due to these limitations, there is a need for a system that can systematically evaluate the credibility of textual claims using machine learning techniques, while maintaining transparency and feasibility for academic implementation.

Therefore, the core problem addressed by this project is the absence of a transparent, scalable, and academically sound system that can automatically assess the credibility of online textual claims using machine learning-based scoring mechanisms.

1.3 Objectives

1.3.1 General Objective

The primary objective of this project is to develop an AI-based fact-checking system that evaluates the credibility of textual claims using machine learning-based scoring techniques.

1.3.2 Specific Objectives

The specific objectives of the project are:

- To analyze textual claims and extract relevant features required for verification
- To design a credibility scoring mechanism using machine learning algorithms
- To compare claims against reliable information sources for consistency analysis
- To classify claims based on their credibility score (e.g., credible, uncertain, or not credible)
- To present verification results in a clear and understandable manner to users

These objectives are aligned with the problem statement and focus on building a feasible and academically valid system.

1.4 Scope and Limitations

1.4.1 Scope of the Project

The scope of this project is limited to text-based fact-checking. The system focuses on verifying short to medium-length textual claims related to news or factual statements. The main scope includes:

- Analysis of textual input provided by users
- Machine learning-based credibility scoring
- Use of algorithmic rules and statistical features for decision-making
- Integration of supportive AI tools for evidence retrieval and explanation

1.4.2 Limitations of the Project

Despite its contributions, the project has the following limitations:

- The system does not verify images, videos, or audio content
- Credibility scoring depends on the quality and availability of online sources
- The machine learning model is trained on limited datasets suitable for academic use
- Real-time social media monitoring is not included
- The system does not guarantee absolute correctness and provides probabilistic assessments

These limitations are acknowledged to ensure realistic expectations and maintain academic integrity.

1.5 Report Organization

The organization of this project proposal is as follows:

- **Chapter 1** presents the introduction of the project, including the problem statement, objectives, and scope of the proposed system.
- **Chapter 2** covers the background study and literature review, focusing on existing fact-checking systems, related research works, their limitations, and the research gap addressed by this project.
- **Chapter 3** discusses the system analysis, which includes problems of the existing system, the proposed system overview, requirement analysis, and feasibility study.

2. Background Study And Literature Review

2.1 Background Study

The term *fact-checking* refers to the process of verifying the accuracy of claims by comparing them against reliable and authoritative sources. With the rapid growth of digital media, automated fact-checking has become an important research area in the fields of Artificial Intelligence (AI), Natural Language Processing (NLP), and Machine Learning (ML).

Automated fact-checking systems generally follow a multi-stage pipeline that includes claim detection, evidence retrieval, claim verification, and result explanation. NLP techniques are used to analyze textual content, extract meaningful features, and represent claims in a structured form. Machine learning algorithms are then applied to classify claims or assign credibility scores based on learned patterns from historical data.

Several approaches have been proposed in academic research, ranging from rule-based systems to supervised machine learning models and neural network-based architectures. While deep learning models have shown promising results, they often require large labeled datasets and lack transparency. As a result, recent studies

emphasize the importance of explainable and interpretable credibility assessment mechanisms, especially in academic and decision-support systems.

This project builds upon these foundational concepts by focusing on machine learning-based credibility scoring combined with transparent decision logic, making the system suitable for academic evaluation and real-world applicability.

2.2 Literature Review

This section reviews existing systems and research related to automated fact-checking and fake news detection. The review highlights their methodologies, advantages, limitations, and the research gap addressed by the proposed system.

2.2.1 Fake News Detection Using Machine Learning

Shu et al. proposed a comprehensive survey on fake news detection that utilizes machine learning techniques such as Logistic Regression, Support Vector Machines (SVM), and Decision Trees to classify news articles based on textual and metadata features [1]. While these methods are computationally efficient, they heavily rely on feature engineering and do not provide detailed credibility explanations.

2.2.2 Neural Network-Based Fact Verification

Wang introduced the LIAR dataset and demonstrated the use of neural networks for fact classification [2]. Although deep learning models achieved reasonable accuracy, the study highlighted challenges related to interpretability and generalization across domains.

2.2.3 Claim Verification with Evidence Retrieval

Thorne et al. presented the FEVER dataset, which focuses on claim verification using evidence extracted from Wikipedia [3]. The system verifies claims by matching them with retrieved evidence sentences. However, such systems depend strongly on structured knowledge sources and may fail when evidence is incomplete or unavailable.

2.2.4 Credibility Assessment Based on Source Reliability

Castillo et al. analyzed credibility in social media content by incorporating source reliability and message features [4]. Their work showed that combining content-based and source-based features improves credibility assessment, but the system lacked automated credibility scoring using machine learning.

2.2.5 Automated Fact-Checking Tools

Existing online tools such as Google Fact Check Explorer and PolitiFact provide verified claims and fact-check reports [5]. These systems rely largely on manual verification and expert analysis, making them accurate but not scalable for large volumes of content.

2.2.6 Hybrid Approaches in Fact-Checking

Recent studies suggest hybrid approaches that combine rule-based logic with machine learning to improve transparency and accuracy [6]. These systems attempt to balance automation with explainability, but many implementations remain experimental or lack real-time applicability.

2.2.7 Limitations of Existing Systems

A review by Zhou and Zafarani emphasized that many fake news detection systems suffer from issues such as data bias, lack of explainability, and over-reliance on black-box models [7]. These limitations highlight the need for systems that provide interpretable credibility assessments.

2.3 Research Gap

Based on the literature review, the following research gaps have been identified:

- Many existing systems focus solely on classification rather than **credibility scoring**
- Heavy reliance on black-box deep learning models limits explainability
- Limited academic systems integrate source reliability, content analysis, and evidence consistency in a unified manner
- Manual fact-checking systems lack scalability, while automated systems lack transparency

2.4 Contribution of the Proposed System

The proposed system aims to address the identified research gaps by:

- Implementing a machine learning-based credibility scoring mechanism
- Combining content-based features with source reliability analysis
- Providing interpretable verification results suitable for academic evaluation
- Reducing dependency on fully black-box AI systems by incorporating transparent algorithmic logic

3. System Analysis

3.1 Requirement Analysis

3.1.1 Functional Requirements

FR ID	Functional Requirement	Description
FR-01	User Input of Claim	The system shall allow users to input textual claims or news statements for verification without requiring authentication.
FR-02	Claim Preprocessing	The system shall preprocess the input text by performing noise removal, tokenization, and text normalization.
FR-03	Claim Feature Extraction	The system shall extract relevant linguistic and statistical features from the input claim for analysis.
FR-04	Evidence Retrieval	The system shall retrieve supporting or contradicting information from predefined reliable online sources.
FR-05	Source Reliability Assessment	The system shall assess the reliability of retrieved sources based on predefined credibility criteria.
FR-06	Credibility Scoring	The system shall compute a credibility score for the claim using a machine learning-based scoring model.

FR-07	Claim Classification	The system shall classify claims as credible, uncertain, or not credible based on the computed credibility score.
FR-08	Explanation Generation	The system shall generate a human-readable explanation describing how the verification result was derived.
FR-09	Result Presentation	The system shall present the verification result, credibility score, and explanation to the user in a clear format.
FR-10	Session-Based Query History	The system shall temporarily store user verification queries during an active session for reference.
FR-11	Error Handling	The system shall handle invalid inputs and unexpected system errors gracefully without crashing.

3.1.2 Non-Functional Requirements

NFR ID	Non-Functional Requirement	Description
NFR-01	Performance	The system shall return verification results within an acceptable response time for standard textual queries.
NFR-02	Scalability	The system shall support an increasing number of verification requests without significant performance degradation.
NFR-03	Usability	The system shall provide a simple and intuitive user interface suitable for users with minimal technical knowledge.
NFR-04	Reliability	The system shall operate consistently and recover gracefully from failures without data corruption.
NFR-05	Security	The system shall ensure protection of user inputs and internal system data from unauthorized access.
NFR-06	Maintainability	The system shall be modular and well-structured to facilitate maintenance and future enhancements.
NFR-07	Explainability	The system shall provide understandable verification outcomes and explanations to improve user trust.
NFR-08	Portability	The system shall be deployable across different platforms with minimal configuration changes.

3.2 Feasibility Analysis

The feasibility analysis evaluates whether the proposed system is practical and achievable within the given constraints of time, resources, and technical capability.

3.2.1 Technical Feasibility

The proposed system is technically feasible as it relies on widely used and well-documented technologies such as Python, machine learning libraries, and web-based interfaces. Machine learning algorithms required for credibility scoring, such as Logistic Regression or similar supervised models, can be trained using publicly available datasets suitable for academic purposes. Text preprocessing and feature extraction techniques are standard practices in Natural Language Processing and do not require specialized hardware.

External AI tools, when used, are limited to evidence retrieval and explanation support, which minimizes dependency on proprietary systems. The system architecture is modular, allowing individual components such as preprocessing, scoring, and result presentation to be developed and tested independently. Therefore, the required technical components are available and manageable within the academic environment.

3.2.2 Economic Feasibility

The system is economically feasible as it does not require significant financial investment. All proposed tools and technologies are either open-source or available under free usage tiers suitable for educational projects. The system does not involve paid hardware, licensing costs, or commercial data sources.

Development is carried out using personal computing resources, and deployment can be done on local or free cloud platforms. As a result, the overall cost of developing and maintaining the system is minimal.

3.2.3 Operational Feasibility

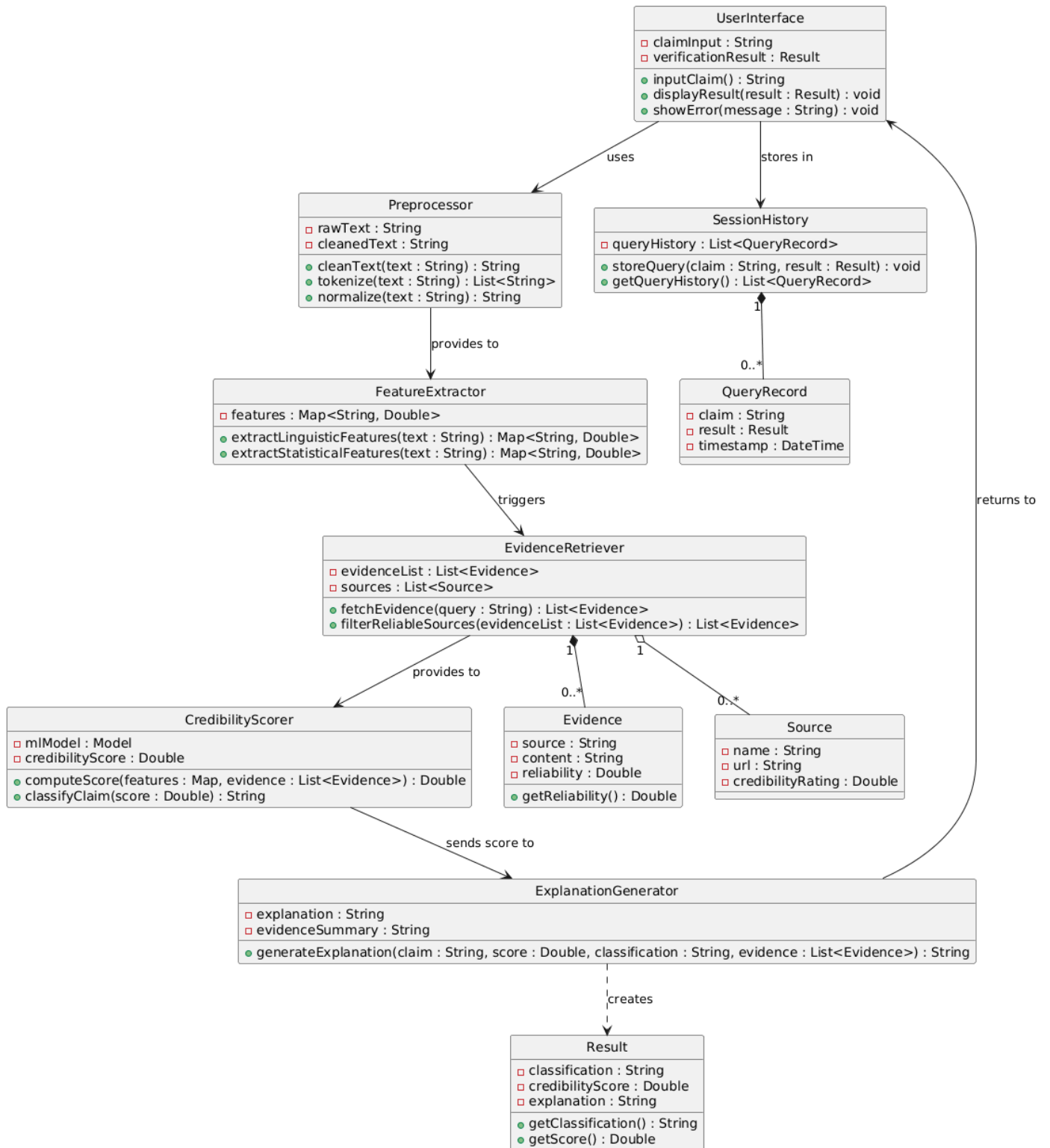
From an operational perspective, the system is designed to be simple and user-friendly. Users can input claims without the need for registration or training. Verification results are presented in a clear and understandable format, making the system accessible to non-technical users.

Administrative operations are minimal and limited to system monitoring and log inspection. Since the system does not alter external data or require complex workflows, it can be operated with minimal supervision.

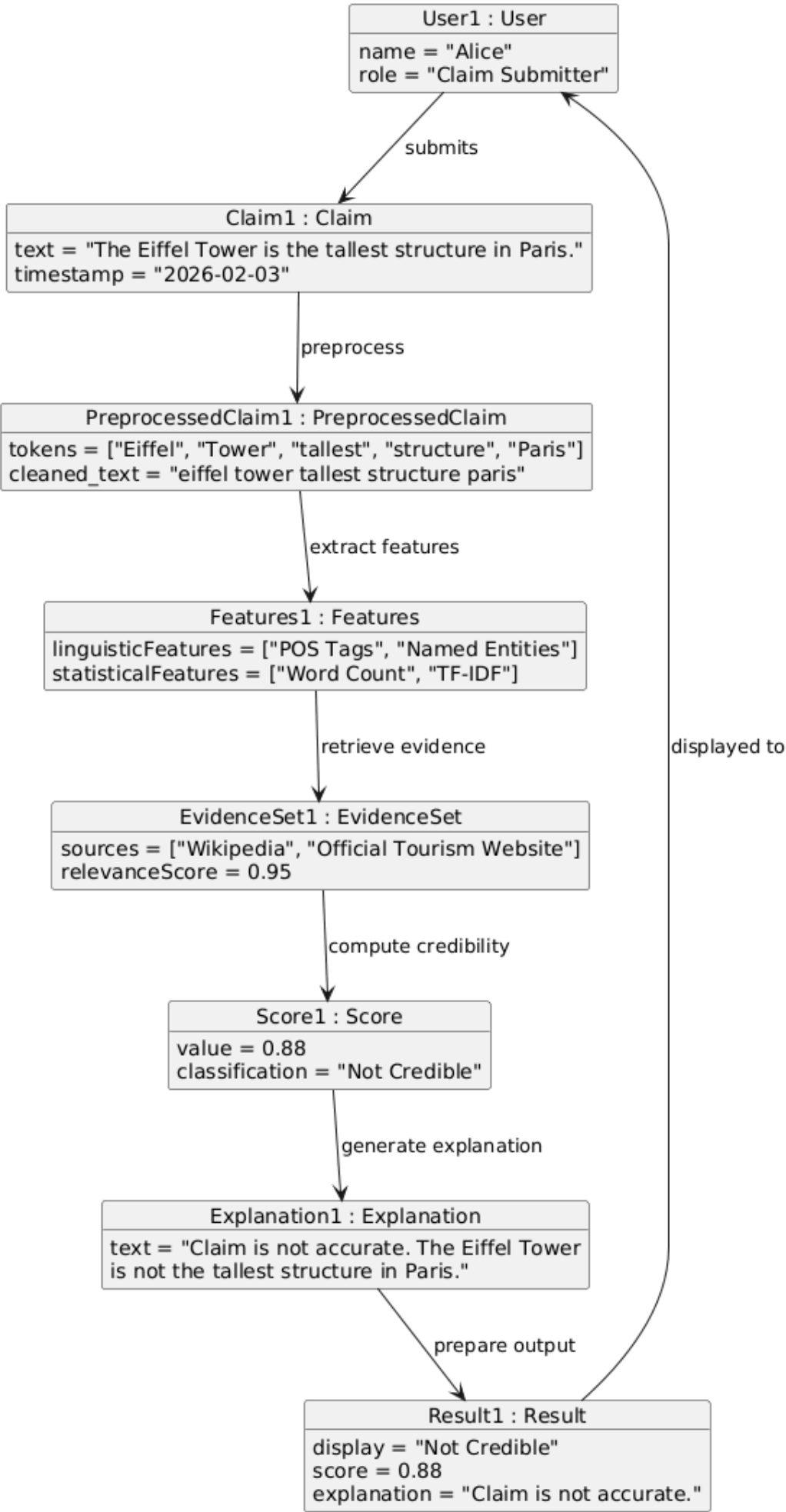
3.3 Analysis Approach

The analysis of the proposed system follows an **Object-Oriented Analysis (OOA)** approach.

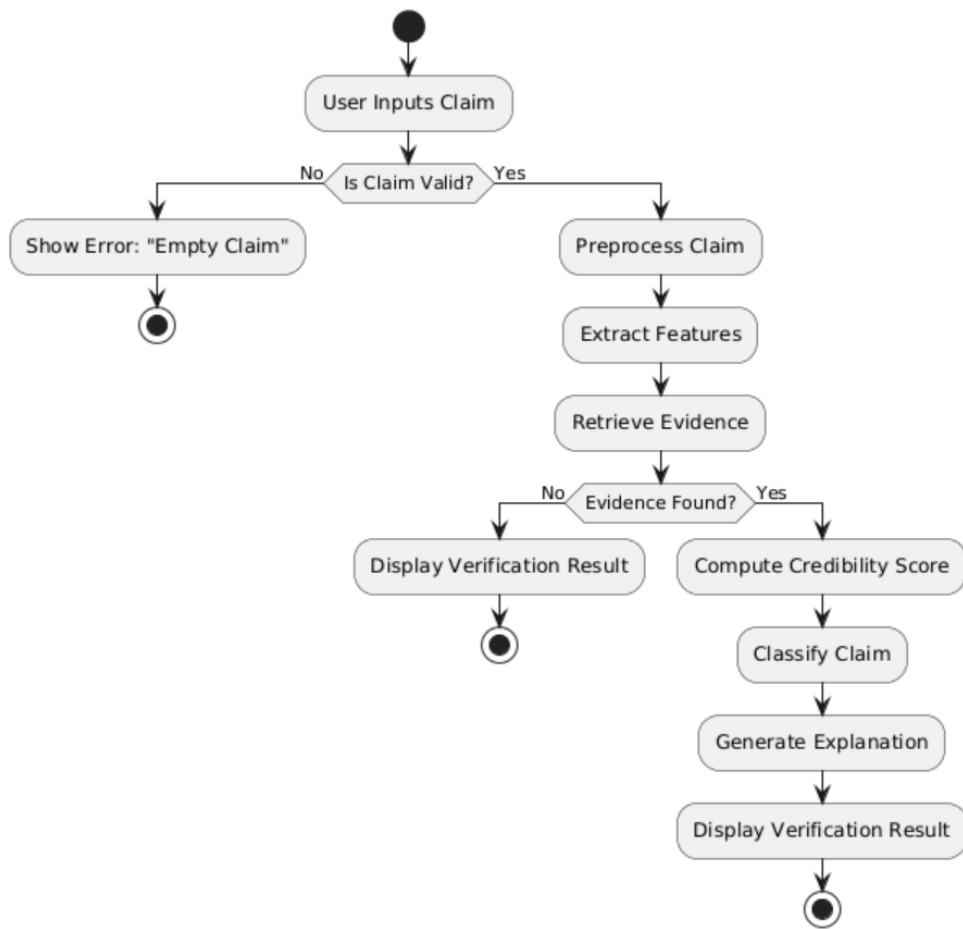
Class Diagram - SatyaScan



Object Diagram - SatyaScan



Activity Diagram - SatyaScan



References

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